Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-2169
Project Name	Import Data using Transform Maps
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution for Import Data using Transform Maps can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited concurrent import jobs	Only a few Transform Map import jobs can run simultaneously; larger teams face queuing delays during peak data loads.	High
2	File size & record volume constraints	Current implementation handles up to ~50k records per job efficiently; larger bulk imports (100k+) cause slower processing and higher memory usage.	High
3	Manual field mapping for complex schemas	Complex multi-object or nested data structures still require significant manual configuration, reducing speed for first-time map setup.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users and sequential import jobs.	Introduce job queue with priority scheduling and multi-tenant isolation so multiple teams can run imports in parallel.
2	Data Storage	Import logs and temporary files stored in standard filesystem / local storage.	Move logs, staging files and audit data to scalable cloud object storage with retention policies and compression.
3	Performance	Single-threaded / limited batch processing for Transform Map execution.	Implement asynchronous batch processing, parallel field transformations, and incremental import mode for large datasets.
4	Security	Basic authentication and role-based access for map configuration.	Add encryption at rest/in transit for import files, detailed audit trails, and fine-grained permissions per Transform Map.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Auto-suggest field mappings using AI / schema similarity and support for more source formats (JSON, XML, API).	Q2 2026	Faster map creation and reduced manual effort for new data sources.
Phase 2	Real-time / streaming import mode and scheduled recurring Transform Map jobs with	Q3 2026	Enable near real-time data sync and automated recurring imports.
Phase 3	Advanced transformation library (custom scripts, multi-object relationships) and visual mapping designer.	Q4 2026	Handle complex enterprise data models with less coding and better usability.
Phase 4	Multi-org / multi-tenant support, full audit & compliance reports, and marketplace of reusable Transform Map templates.	Q1 2027	Enterprise readiness, easier adoption across teams, and accelerated onboarding of new use cases.